

ISO 26262

1. Introduction

a. What is Functional Safety

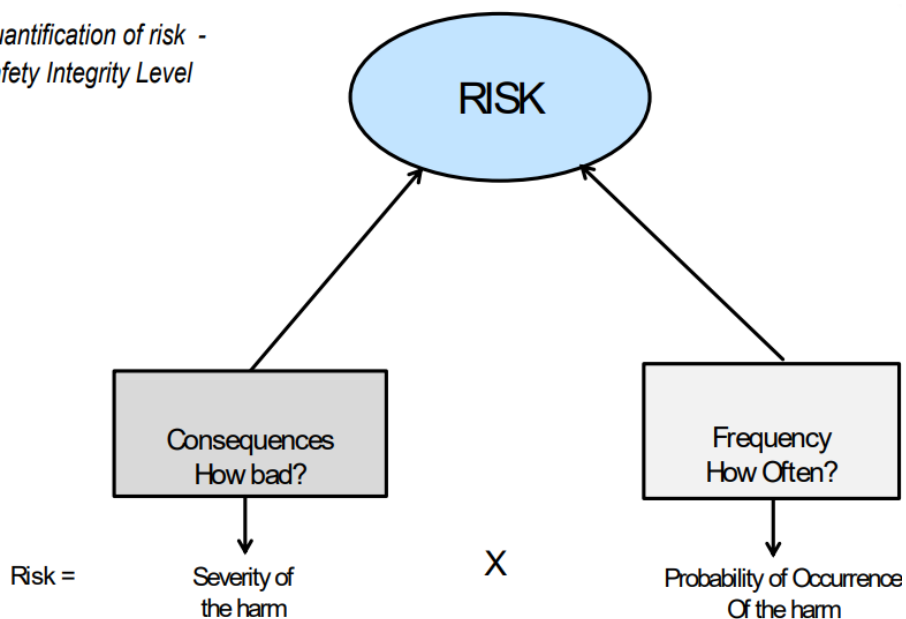
- Functional safety ensures systems work correctly in response to inputs.
- Prevent hazards (mối nguy hiểm) caused by system failures or malfunctions.
- Provides risk reduction to achieve equipment safety.
- Mục tiêu: giảm rủi ro xuống mức chấp nhận được bằng Safety Mechanisms.

b. ISO 26262

- It concerns the functional safety of automotive systems.
- It addresses unintended activations of functions.
- It ensures operators know when functions aren't activated.
- It requires systems to enter a safe state if a critical component fails.

c. Risk Reduction

*Quantification of risk -
Safety Integrity Level*



d. Safety Integrity Level

- Safety Integrity Level (SIL) is a way to measure and classify risk in safety systems.
- It indicates how reliably a system must perform its safety functions.
- SIL is determined by factors like severity, likelihood, and controllability of hazards.
- In automotive, risk is classified as ASIL (A to D), **with D being the strictest**.

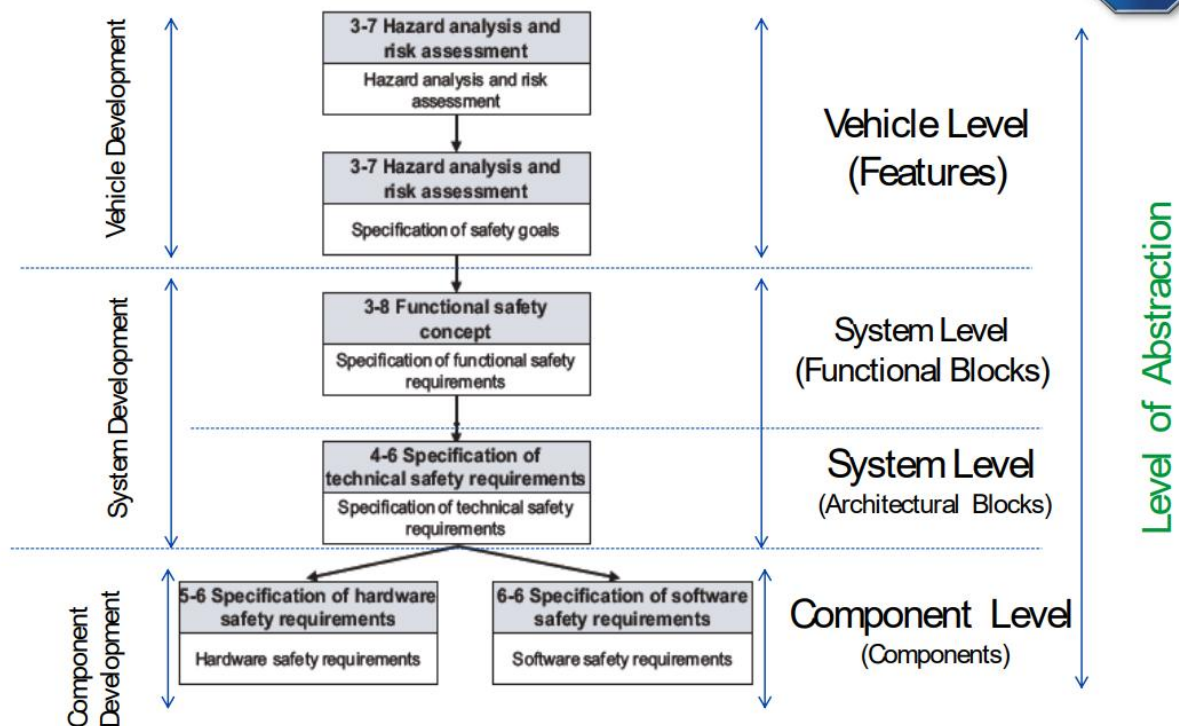
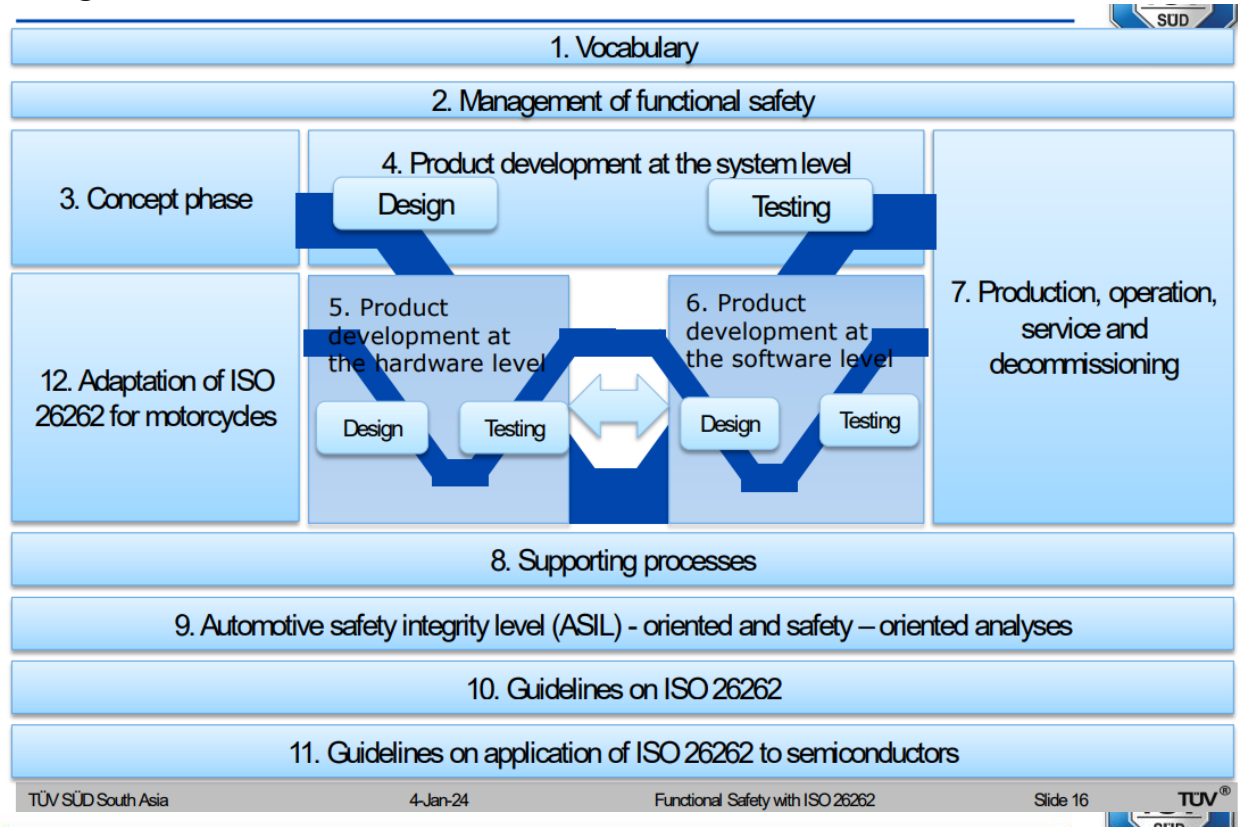
e. Random and Systematic Failures

Aspect	Random Failures	Systematic Failures
Cause	Spontaneous, unpredictable events	Design, process, or implementation faults
Occurrence	At any time during operation	Can occur at any product lifecycle stage
Example	Hardware component failure	Software bug, design error
Detection	Hard to predict, statistical analysis	Can be detected and fixed
Elimination	Cannot be fully eliminated	Can be eliminated after detection

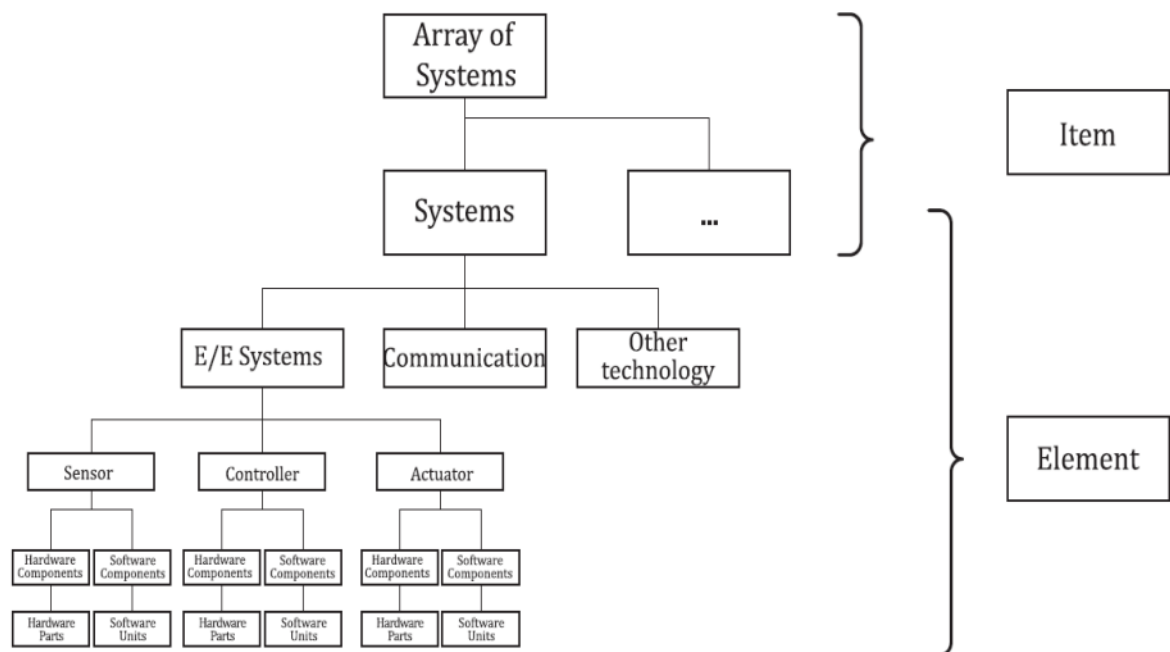
f. Features

- The new ISO 26262-2018 standard replaces the previous 2011 version
- Introduces several key changes.
- Covers requirements for more vehicle types (like trucks and motorcycles)
- Expands vocabulary, adds more detailed objectives, and includes measures for confirming objectives.
- Address safety anomaly management, references cyber security, updates hardware metrics, and provides guidance on software, fault tolerance, semiconductors, and dependent failures.
- Restructures all parts for better clarity and completeness.

g. Overview of ISO 26262 Standards



h. Item, Element and Component



i. Some definitions

ASIL:	Automotive Safety Integrity Level
Base Failure Rate (BSR)	Fault rate of a hardware element provided by manufacturer FIT
OEM	Original Equipment Manufacturer
T&B	Trucks, Buses, trailers and semi-trailers
Calibtion data	- Software-based data used to adjust and fine-tune the operation of a system or device - Stored in EEPROM or flash - Control how the software behaves - Ex: lờ idle speed, country code, right-hand steering
Development Interface Agreement (DIA)	Contract between customer and supplier outlining responsibilities, evidence review, and exchange of work products during development.
Fault	Abnormal condition that can cause an element to fail
Failure	Termination of the ability of an element
Random HW failure	- Sự hỏng hóc của linh kiện phần cứng tại bất kỳ thời điểm bất kỳ, do các yếu tố như lão hóa, hao mòn, hoặc môi trường - Không thể dự đoán trước. - Không liên quan đến thiết kế hay quá trình sản xuất - Xuất hiện ngẫu nhiên trong quá trình vận hành thực tế.
Cascading failure	- Sự cố dây chuyền

Common Cause failure (CCF)	- Hai hay nhiều thành phần hoặc hệ thống bị hỏng cùng lúc do cùng một nguyên nhân - Điều này khiến các biện pháp dự phòng mất hiệu lực
Dependent failure	- Include common cause failures and cascading failures - Các hỏng hóc không độc lập về mặt thống kê, nghĩa là xác suất xảy ra đồng thời của các hỏng hóc này không bằng tích của xác suất xảy ra của từng hỏng hóc nếu chúng được coi là độc lập. - Khi một phần tử bị hỏng, nó có thể gây ra hoặc làm tăng khả năng các phần tử khác cũng bị hỏng.
Detected fault	- Lỗi được phát hiện bởi safety mechanism
Dual point failure	- Xảy ra đồng thời hai lỗi độc lập hoặc có liên quan trong một hệ thống, dẫn đến việc mất an toàn hoặc chức năng quan trọng.
Latent fault	- Lỗi tiềm ẩn - Không được phát hiện ngay lập tức bởi các cơ chế giám sát hoặc kiểm tra hiện tại. - Chỉ được phát hiện khi có một sự kiện hoặc kiểm tra riêng biệt xảy ra, hoặc khi nó kết hợp với một lỗi khác gây ra hậu quả nghiêm trọng.
Single point fault	- Xảy ra ở một điểm duy nhất. - Không có biện pháp phát hiện hoặc dự phòng

	- Khi lỗi này xảy ra sẽ dẫn đến nguy hiểm ngay lập tức.
Multiple point fault	- hai hoặc nhiều phần tử (elements) trong hệ thống bị hỏng cùng lúc - Những lỗi này có thể độc lập (independent) hoặc phụ thuộc (dependent), như lỗi do nguyên nhân chung (common cause failure) hoặc lỗi dây chuyền (cascading failure).
Safe fault	-
Perceived fault	- Lỗi nhận biết được bởi người dùng
Residual fault	- Trong một phần tử phần cứng, có một số lỗi đã được cơ chế phát hiện, nhưng vẫn còn một phần lỗi chưa được kiểm soát và nó có thể gây nguy hiểm. - Residual Risk: phần rủi ro còn lại
Diagnostic Coverage	- Mức độ mà hệ thống có thể phát hiện hoặc kiểm soát các lỗi xảy ra trong phần cứng nhờ các biện pháp bảo vệ.
Fault detection time interval (FDTI)	- Từ khi lỗi xảy ra đến khi được phát hiện
Fault reaction time interval (FRTI)	- Từ khi phát hiện lỗi đến khi đạt đến safe state hoặc emergency operation
Fault handling time interval (FHTI)	- FDTI + FRTI
Fault tolerance	- Khả năng mà hệ thống có thể tiếp tục hoạt động bình thường (hoặc ở mức chấp nhận được) dù xảy ra lỗi.
Fault tolerance time interval (FTTI)	- Thời gian tối thiểu từ khi lỗi xảy ra trên một thiết bị cho đến khi có thể xảy ra một sự kiện nguy hiểm, trong trường hợp các cơ chế an toàn không được kích hoạt. - Tức thời gian “chịu đựng được” của hệ thống trước khi dẫn đến tình huống nguy hiểm nếu lỗi không được xử lý kịp thời.
Freedom from interference	- Là sự vắng mặt của các lỗi dây chuyền (cascading failures) giữa hai / nhiều thành phần mà có thể dẫn đến việc vi phạm yêu cầu an toàn. - Ví dụ: Một lỗi trong hệ thống điều khiển phanh không được phép lan sang hệ thống điều khiển túi khí.
Exposure (Tiếp xúc)	- Là trạng thái trong một tình huống vận hành có thể trở nên nguy hiểm nếu xảy ra trùng hợp với chế độ lỗi đang được phân tích. - Ví dụ: Một máy bay đang trong quá trình cất cánh là trạng thái "tiếp xúc" với rủi ro cao nếu hệ thống điều khiển động cơ gặp lỗi.
Independent failures	- Các lỗi xảy ra trong một hệ thống hoặc giữa các thành phần nhưng hoàn toàn không liên quan đến nhau. - Xác suất xảy ra lỗi ở một thành phần không ảnh hưởng đến xác suất xảy ra lỗi ở một thành phần khác.

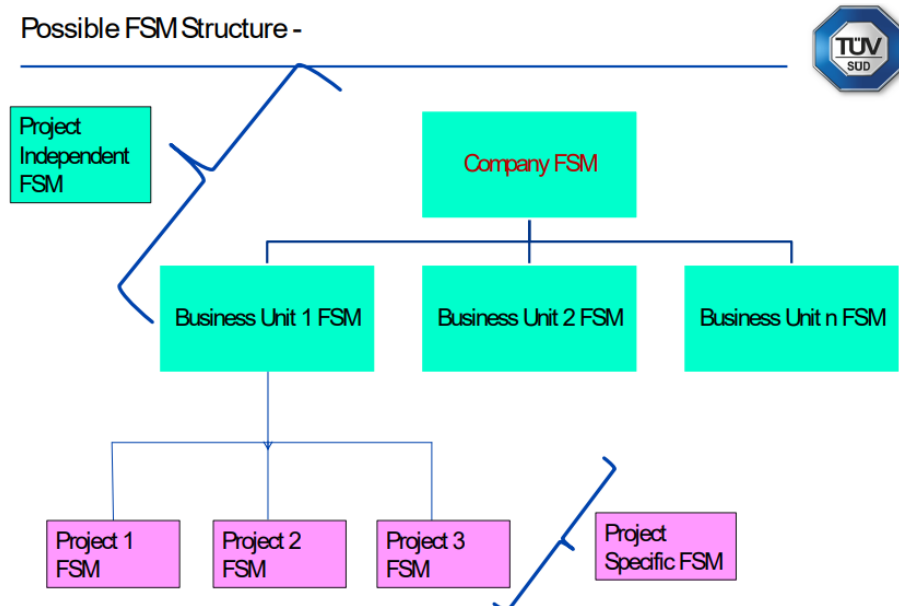
j. Note:

- — “++” indicates that the method is highly recommended for the identified ASIL;
- — “+” indicates that the method is recommended for the identified ASIL;
- — “o” indicates that the method has **no recommendation** for or against its usage for the identified ASIL.

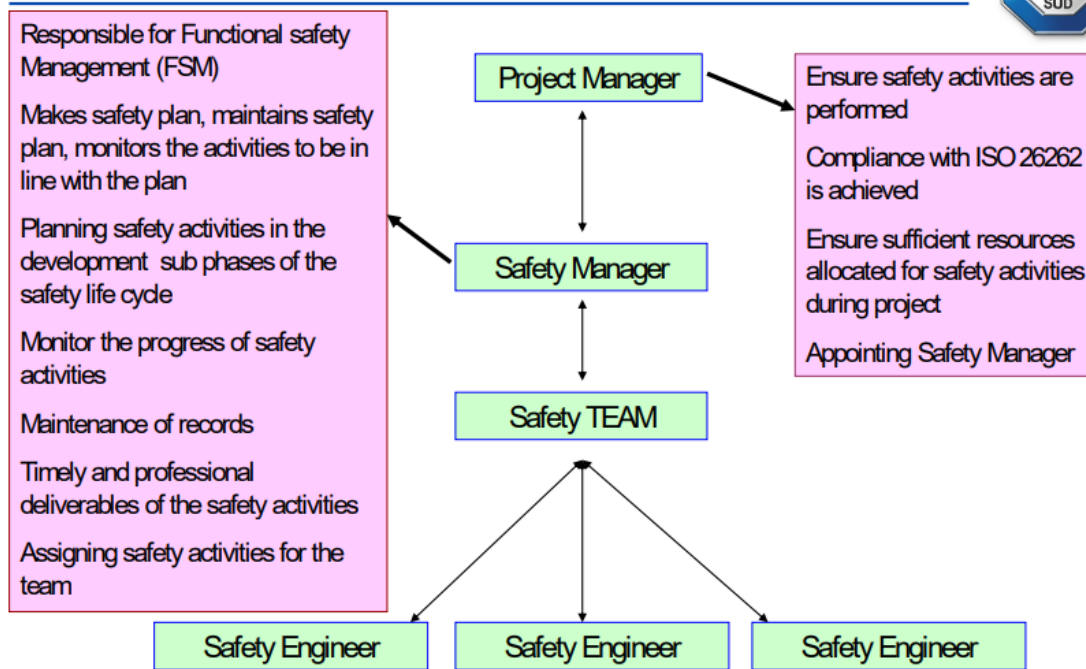
2. Functional Safety Management (FSM)

a. Possible FSM Structure

Possible FSM Structure -



b. Roles and responsibilities in safety management



c. Reuse of existing element

- Base on an evaluation of HW elements
- Base on qualification of SW components
- Base on proven in use argument
- As a Safety Element out of Context

d. Safety plan

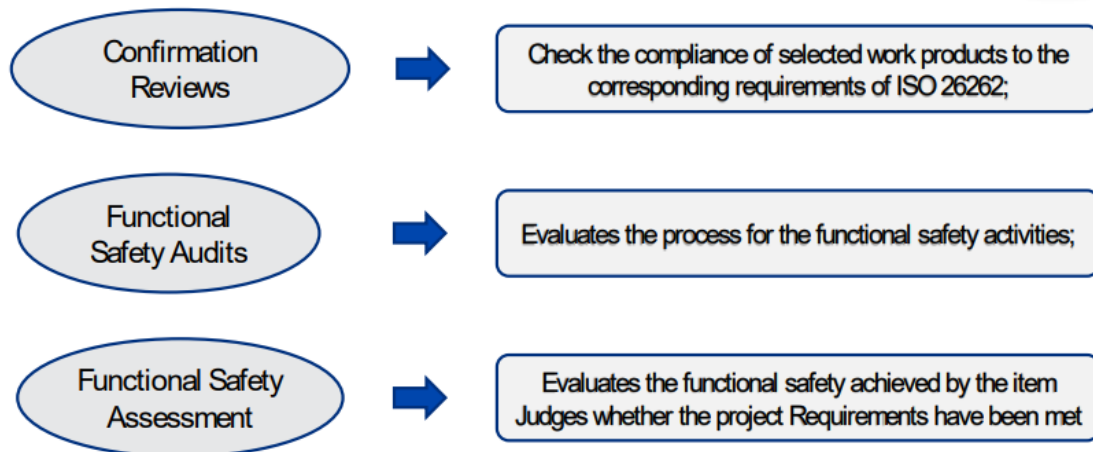
Prepare by safety manager, approved by project manager

e. Safety case

- Tài liệu được phát triển để chứng minh rằng một hệ thống hoặc sản phẩm đáp ứng các yêu cầu về functional safety trong vòng đời của nó.
- Gồm các lập luận, bằng chứng, và dữ liệu để hỗ trợ cho tuyên bố rằng sản phẩm hoặc hệ thống đó là an toàn.
- Phát triển dựa trên safety plan
- Cần được xây dựng và cập nhật dần dần
- không phải là hoạt động chỉ thực hiện vào cuối safety life cycle.

f. Confirmation Measures

- Confirmation Reviews, Functional Safety Audits, Functional Safety Assessment.



g. Level of Independence

Level	Description
I0	Confirmation measure should be performed If performed, it must be by a different person than the one who created the work product.
I1	Confirmation measure must be performed by a different person than the one who created the work product.
I2	Confirmation measure must be performed by a person independent from the team that created the work product (not reporting to the same direct superior).
I3	Confirmation measure must be performed by a person who is independent in terms of management, resources, and release authority, from a different department than the one responsible for creating the work product.

h. Confirmation measures with level of Independence

Confirmation Measure	Apply to ASIL
Confirmation review of the impact analysis at the item level	QM, A, B, C, D: I3
Confirmation review of the hazard analysis and risk assessment	QM, A, B, C, D: I3
Confirmation review of the safety plan	A, B: I1 C: I2 D: I3
Confirmation review of Functional Safety Concept	A, B: I1 C: I2 D: I3
Confirmation review of Technical Safety Concept	A, B: I1

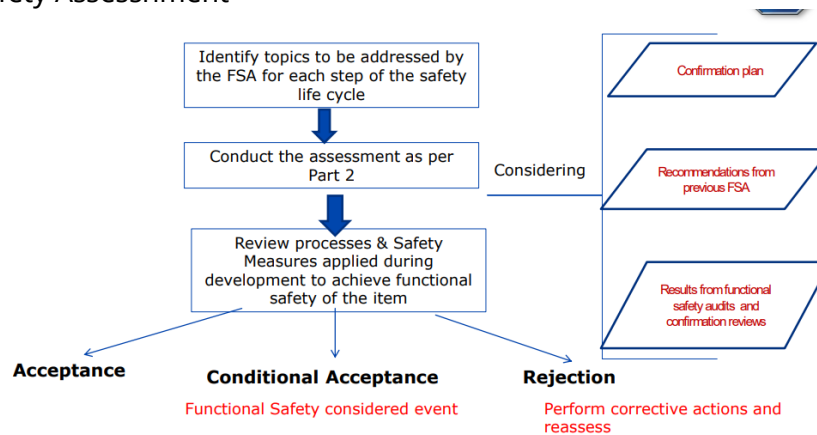
	C: I2 D: I3
Confirmation review of the Intergration and Test Strategy	A: I0 B: I1 C, D: I2
Confirmation review of Safety Validation Specs	A: I0 B: I1 C, D: I2
Confirmation review of Safety Analysis and Dependent Failures	A, B: I1 C: I2 D: I3
Confirmation review of the Safety Case	A, B: I1 C: I2 D: I3
Confirmation review of the Functional Safety Audit	B: I0 C: I2 D: I3
Confirmation review of the Functional Safety Assessment	B: I0 C: I2 D: I3

i. Defination

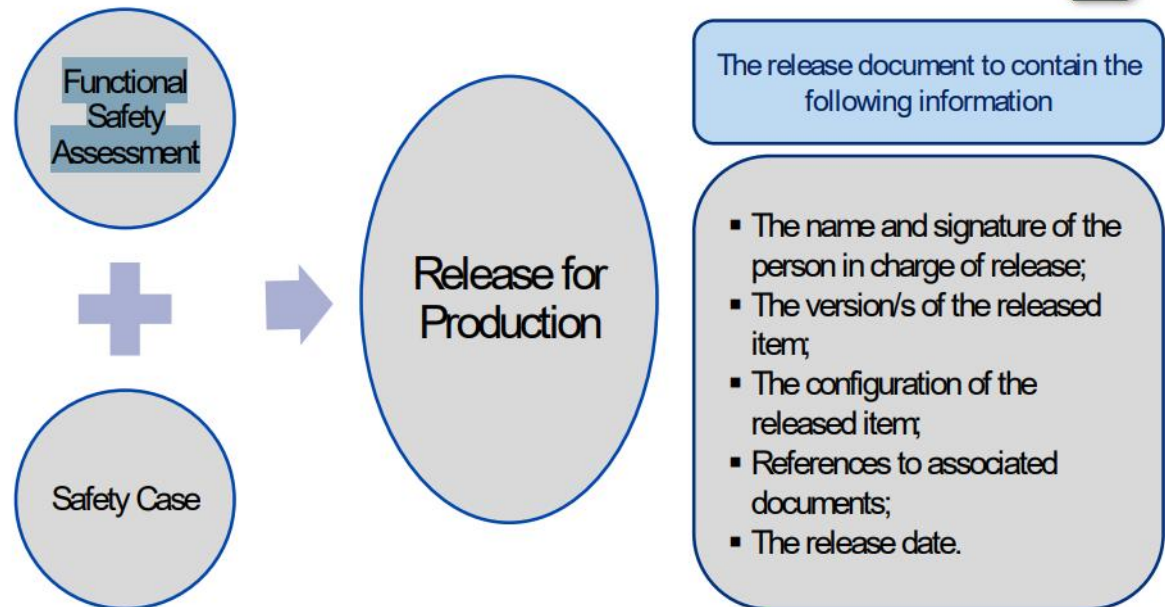
Confirmation Review

Functional Safety Audit

Functional Safety Assessment

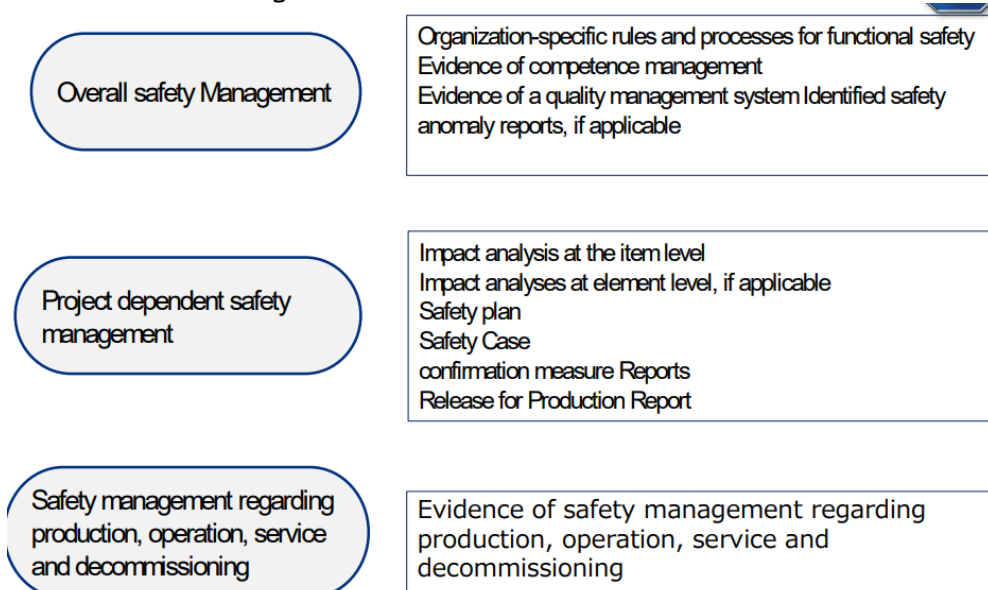


j. Release for production



Functional Safety Management:

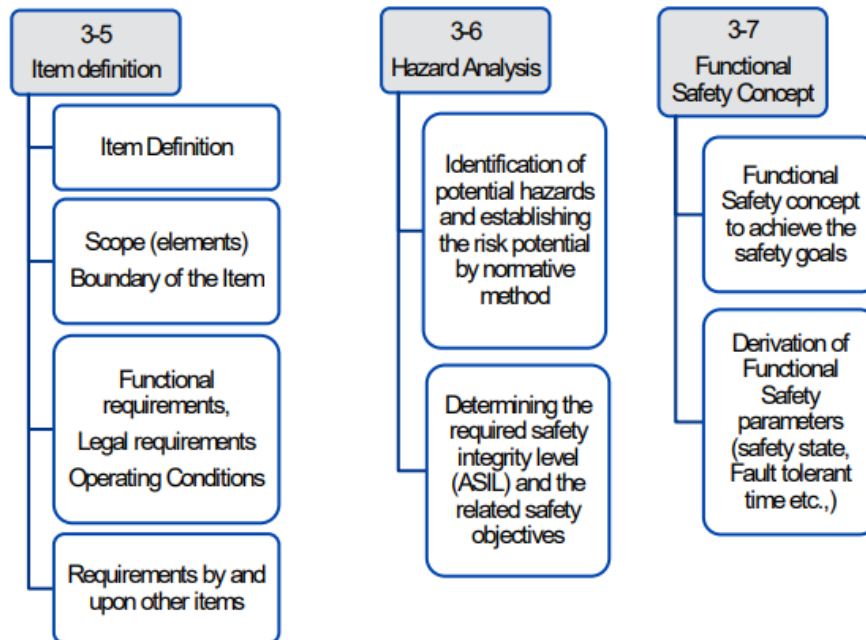
- Overall safety Management
- Project dependent safety
- Safety management regarding production, operation, service and decommissioning



3. Concept phase

a. Steps

Item Definition -> Hazard Analysis -> Functional Safety Concept



b. Item definition

Boundary Interfaces Assumptions	Element of item Giả định liên quan đến ảnh hưởng của hành vi item đối với xe; chức năng của item chức năng của những item khác mà nó đang cần đến; kịch bản vận hành ảnh hưởng đến chức năng của hạng mục.
Requirement of item	- Legend requirement (internal và external standard) - Functional behavior at vehicle level - Required quality, performance and availability of functionality - Constraints (ràng buộc liên quan) - Hậu quả tiềm ẩn, (potential consequences) gồm các dạng hỏng đã biết và rủi ro an toàn - the capabilities of the actuators

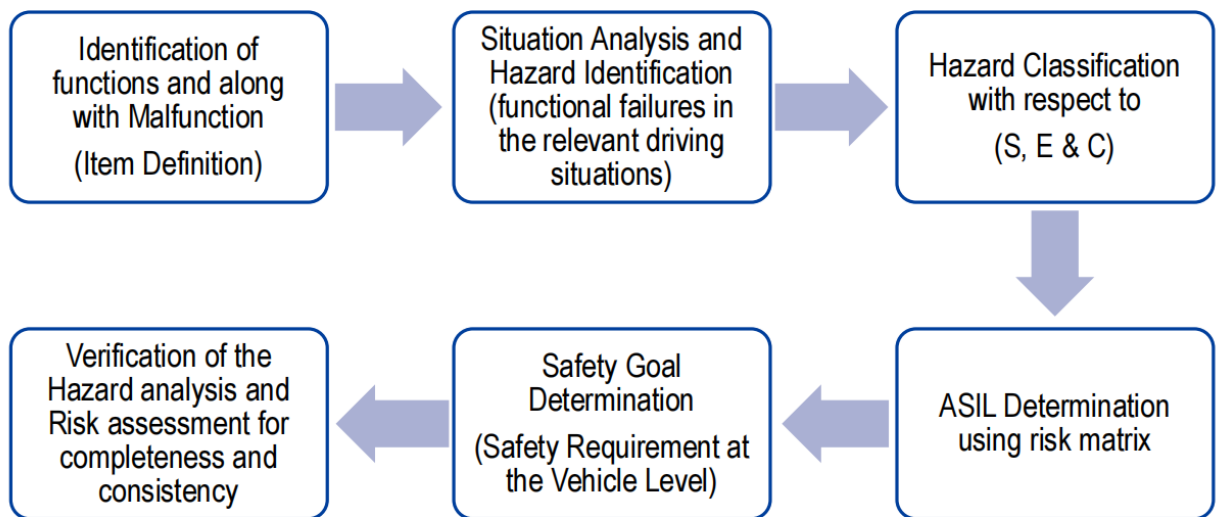
c. Hazard Analysis (HA) and Risk Assessment (RA)

Purpose:

- Identify and classify hazardous events caused by item malfunctions.
- Define safety goals and their ASILs to prevent or mitigate these hazardous events and avoid unreasonable risk.

Required input:

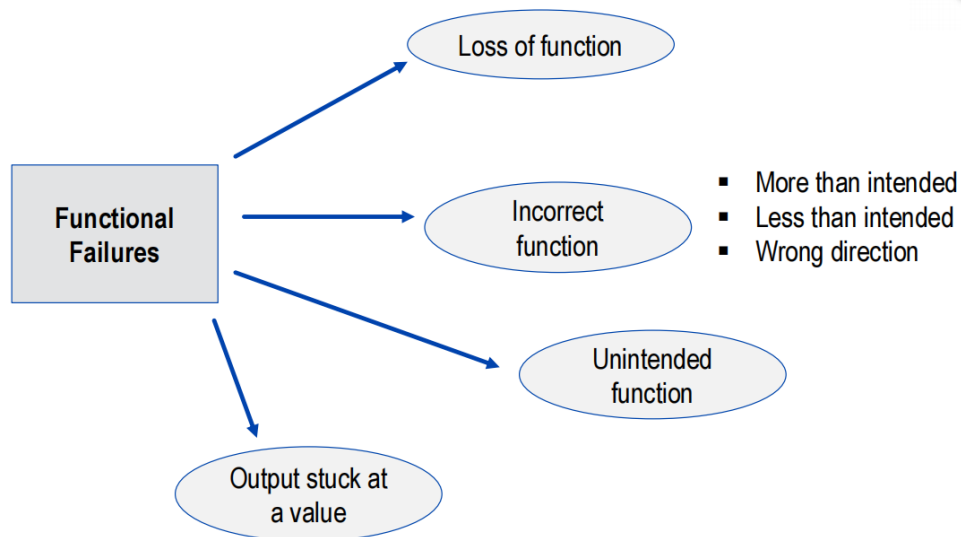
- Item Definition
- Relevant information about other items (from external sources), if available



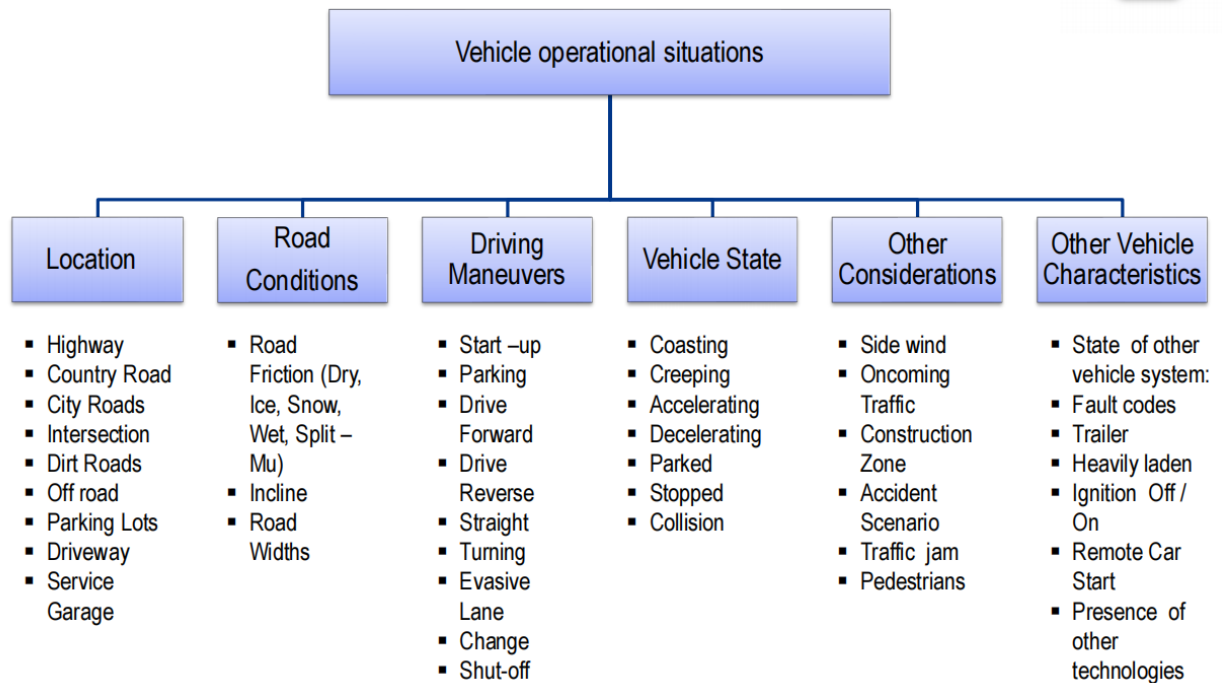
d. Hazard Identification

Check document

e. Possible Functional Failures



Potential operational vehicle situations (from SAE J2980)



Location	• Highway • City Road / Country Road • Intersection – Ngã tư • Off-road / Dirt Roads – Đường đất / địa hình • Parking Lots / Driveway • Service Garage – Xưởng bảo dưỡng
Road Conditions	• Road friction (dry, wet, snow, ice) • Incline – Độ dốc • Road widths – Độ rộng đường
Driving Maneuvers Các thao tác lái xe	- Start-up - Parking - Drive forward / reverse - Straight / turning - Lane change – Chuyển làn - Evasive maneuver – Né tránh - Shut-off – Tắt máy
Vehicle State	• Coasting – Xe trôi • Creeping – Xe bò chậm (thường trong tắc đường) • Accelerating / decelerating – Tăng tốc / giảm tốc • Parked / Stopped – Đỗ / dừng • Collision – Va chạm
Other Considerations	• Side winds, Oncoming traffic, Road construction, Accidents, Traffic jams, Pedestrians

Other Vehicle Characteristics	• Fault codes – Mã lỗi hệ thống • Trailer – Có kéo rơ-moóc • Heavily laden – Chở nặng • Ignition on/off, remote start – Bật/tắt máy, khởi động từ xa • Presence of other technologies – Có thêm công nghệ khác trong xe
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f. Hazard Classification

Hazard severity is classified into four levels (S0–S3), based on injury severity to passengers, drivers, pedestrians, or others.

S0	No injuries AIS 0, hoặc dưới 10% khả năng xảy ra AIS 1–6.
S1	Light / Moderate injuries >10% khả năng AIS 1–6
S2	Life-threatening injuries (survival probable) >10% khả năng AIS 3–6
S3	Life-threatening (survival uncertain) >10% khả năng AIS 5–6.

AIS Scale

- AIS 0 – No Injury
- AIS 1 – Minor Injuries
- AIS 2 – Moderate Injuries: rách da sâu, chấn động nhẹ (bất tỉnh < 15 phút), gãy xương nhẹ.
- AIS 3 – Serious (Non-life-threatening): gãy sọ không tổn thương não, trật cột sống không tổn thương tủy, nhiều xương sườn gãy.
- AIS 4 – Severe (Life-threatening, survival probable): gãy sọ nặng, chấn thương cột sống, bất tỉnh đến 12 giờ.
- AIS 5 – Critical (Survival uncertain): cực kỳ nghiêm trọng, đe dọa tính mạng: tổn thương não, tim, nội tạng, bất tỉnh > 12 giờ.
- AIS 6 – Maximum / Fatal: Tử vong

Probability of Exposure

Đó là mức độ thường xuyên mà người lái / xe sẽ gặp tình huống giao thông có thể dẫn đến nguy hiểm, nếu lỗi xảy ra.

E0 – Incredible

E1 – Very Low Probability

E2 – Low Probability: <1%

E3 – Medium Probability: 1%–10%

E4 – High Probability: >10%

Controllability

mức độ người lái hoặc người tham gia giao thông có thể tránh được tai hại nếu sự cố xảy ra.

Class	C0	C1	C2	C3
Description	Controllable in general	Simply Controllable	Normally Controllable	Difficult to Control or uncontrollable
	Controllable in General	99% or more of all drivers or other to traffic participants are usually able to avoid a specific harm	90% or more of all drivers or other to traffic participants are usually able to avoid a specific harm	Less than 90% of all drivers or other traffic participants are usually able or barely able to avoid specific harm

g. ASIL classification

Severity class	Exposure class	Controllability class		
		C1	C2	C3
S1	E1	QM	QM	QM
	E2	QM	QM	QM
	E3	QM	QM	A
	E4	QM	A	B
S2	E1	QM	QM	QM
	E2	QM	QM	A
	E3	QM	A	B
	E4	A	B	C
S3	E1	QM	QM	A ^a
	E2	QM	A	B
	E3	A	B	C
	E4	B	C	D

h. Safety goals

Mỗi hazardous event trong HARA phải có một Safety Goal.

ASIL của hazardous event được gán trực tiếp cho Safety Goal tương ứng.

Các Safety Goal giống nhau có thể gộp thành một.

Khi gộp, SG mới lấy ASIL cao nhất trong nhóm.

Nếu Safety Goal đạt được bằng cách đưa hệ thống vào một safe state, thì phải xác định rõ safe state đó.

Các thuộc tính (atribute)

- **ASIL of SG** – Mức ASIL gán cho SG.

- **Safe State** – Trạng thái an toàn mục tiêu.
- **Fault Tolerance Time Interval (FTTI)** – Thời gian từ khi lỗi xảy ra đến khi gây nguy hiểm.
- **Warning Concept** – Cách cảnh báo người lái.
- **Degradation Concept** – Cách giảm chức năng để giữ an toàn (ví dụ: limp-home).

i. T&B Variances in HARA

Cần xem xét: base vehicle, vehicle operation (ko tải, đầy tải, đô thị, miền quê), configuration (có rơ-móc hay không, loại thùng hàng, độ cao trọng tâm)

T&B variances = thay đổi loại xe, cấu hình xe, và cách vận hành → làm thay đổi S-E-C → thay đổi ASIL.

j. Functional safety requirement (FSR)

Mục tiêu là đảm bảo: Fault Tolerant Time Interval – FTTI đúng quy định, và giảm thiểu hazardous events resource conflict.

- 1 Safety Goal → ít nhất một FSR.
- Một FSR có thể áp dụng cho nhiều Safety Goals.
- Khi xác định FSR, cần xét: Operating modes, Fault handling time, Safe states, Emergency operations, Redundancy (ví dụ: fault tolerance).

FSR Format

- **Unique ID** (mã định danh duy nhất).
- **ASIL** (thừa hưởng từ Safety Goal).
- **Element / component** liên quan trong kiến trúc.
- **Verification method**
- **Safety Goal Mapping** (liên kết với SG nào).

Ví dụ:

Khi EPS mất trợ lực, xe phải chuyển sang chế độ lái cơ trong <500 ms, mô-men lái không vượt 15 Nm, và cảnh báo người lái trong <1 s.

FSR-EPS-001

Title: Maintain Basic Steering Ability in Case of Torque Assist Failure

ASIL: ASIL C (From Safety Goal SG-EPS-001)

Safety Goal Mapping:

SG-EPS-001 – *Prevent loss of steering control due to EPS malfunction.*

Requirement Statement (Functional Safety Requirement):

The EPS system shall ensure that, in the event of a loss of torque assist, the driver can continue to mechanically control the steering with a steering wheel torque not exceeding 15 Nm.

Rationale:

Giảm nguy cơ **loss of control** khi EPS fails, đảm bảo khả năng điều khiển cơ học của người lái.

Operating Modes Covered:

- Speed > 5 km/h
 - Normal driving, highway, low-speed maneuvering
 - All steering demand ranges
-

Fault Handling / Timing:

- The system shall detect loss-of-assist within < **100 ms**.
 - Within the **Fault Tolerant Time Interval (FTTI)** of **500 ms**, the EPS shall transition to **manual fallback mode**.
-

Safe State:

- “Mechanical fallback steering mode” activated.
 - Torque assist = 0 Nm, but steering column fully functional mechanically.
-

Warning Concept:

- Driver shall be informed within < **1 s** via a steering warning indicator (ISO icon, amber).
-

Degradation Concept:

- Assist function degraded → vehicle remains steerable mechanically; no self-centering torque applied.
-

Verification Method:

- Functional test on vehicle (torque measurement).
 - Software fault injection test verifying detection time.
 - Hardware-in-loop (HIL) simulation for fallback timing.
 - Requirements review.
-

k. Work Products at the End of Concept Phase

Item Definition

Hazard Analysis & Risk Assessment (HARA) + Safety Goals

Verification of HARA

Functional Safety Concept (FSC)

Verification of Functional Safety Requirements (FSR)

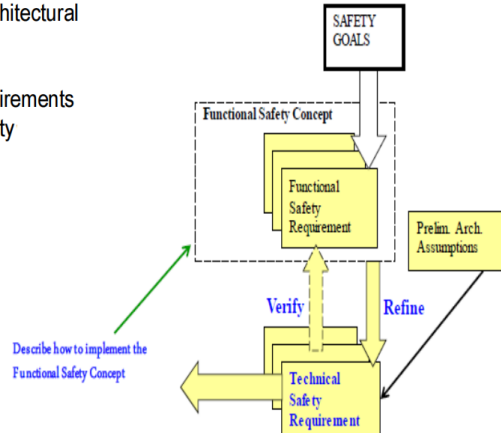
4. Product development at system level

a. Phase Model

FSC → TSC → HW/SW Development → Integration → Validation.

Aim: Develop the Technical safety requirements to refine the Functional safety requirements considering the preliminary architectural assumptions

Verify the technical safety requirements comply with the functional safety requirements



- Technical Safety Concept (TSC): tập hợp TSR + System Architecture, kèm lý do giải thích vì sao kiến trúc đó đủ an toàn để đáp ứng Safety Goals từ HARA.
- Technical Safety Requirements (TSR): xác định: Cách triển khai an toàn trong kiến trúc, Cách phát hiện/giảm thiểu lỗi ẩn, Cách tránh lỗi hệ thống, Yêu cầu vận hành & bảo trì an toàn
- System Architectural Design: Là phương án kiến trúc hệ thống được chọn để hiện thực hóa TSC. Phải đáp ứng TSR và các yêu cầu khác như performance, packaging, thermal.

b. Technical Safety Concept (TSC)

Là tập hợp TSR + System Architecture, cùng “rationale” giải thích vì sao kiến trúc đó đáp ứng yêu cầu an toàn.

Dùng để chuyển từ FSC → mức kỹ thuật (HW/SW).

Mục tiêu:

- Chuyển FSR → TSR.
- Xác định Safety Mechanisms.
- Xây System Architectural Design.

c. Technical Safety Requirements (TSR)

Yêu cầu an toàn kỹ thuật, chi tiết cách implement FSR xuống hệ thống.

Input

- Functional safety concept, functional safety requirement
- System Architectural Design
- Requirement from other safety relevant item

Prefer

- HARA analysis report
- Item Definition document

Safety Mechanisms

TSR phải xác định:

- Fault detection
- Fault indication

- Fault control
- Self-monitoring
- External fault detection (ECUs khác, nguồn, communication)
- Warning & degradation concept
- Prioritization / arbitration logic
- Safe state logic
- Tests trước khi lái (pre-drive), sau khi lái (post-drive), maintenance tests

d. TSR – Avoidance of the latent fault

Để xác định khả năng phát hiện multiple-point fault, cần xem xét các tham số sau:

- Độ tin cậy của linh kiện và vai trò của linh kiện đó trong kiến trúc hệ thống.
- Probability of Exposure đối với các sự kiện nguy hiểm tương ứng.
- Giá trị mục tiêu định lượng được chỉ định cho xác suất tối đa mà mỗi safety goal có thể bị vi phạm do lỗi ngẫu nhiên của phần cứng.

ASIL được gán cho safety goal liên quan.

- Các cơ chế an toàn nhằm ngăn ngừa lỗi kép (dual-point faults) trở thành lỗi tiềm ẩn phải được xây dựng theo quy định sau:
- ASIL B cho các Technical Safety Requirements có ASIL D.
- ASIL A cho các Technical Safety Requirements có ASIL B và C.
- Engineering judgment (đánh giá kỹ thuật) cho các Technical Safety Requirements có ASIL A.

e. TSR format

- TSR ID no
- TSR description
- ASIL of the requirement (either inherited from FSR or ASIL Decomposition)
- Classification as to HW or SW or HW/SW related requirement
- Architectural element to which it belongs to
- Verification criteria
- Mapping to FSR

f. System architecture design and technical safety concept

System architecture design:

Các yêu cầu chính (Khi triển khai phải đảm bảo)

- Implement đầy đủ TSR
- Verifiable (thiết kế phải kiểm chứng được)
- HW/SW suitability (phần cứng & phần mềm phù hợp để đạt safety)
- Testable (cho phép chạy test ở system integration)

Base on: Item Definition, Functional Safety Concept, prior architectural design

Phải thực hiện đầy đủ các Technical Safety Requirements

g. Measures of Avoidance of Systematic Failures

Measures of Avoidance of Systematic Failures



Methods		ASIL			
		A	B	C	D
1	Deductive analysis ^a	0	+	++	++
2	Inductive analysis ^b	++	++	++	++
^a Deductive analysis methods include FTA, reliability block diagrams, Ishikawa diagram.					
^b Inductive analysis methods include FMEA, ETA, Markov modelling.					

h. Hardware–Software Interface (HSI)

Phải định nghĩa đầy đủ:

- Operating modes (init, normal, degraded, safe)
- Memory map, registers, interrupts
- Timing / triggers
- Data integrity
- Diagnostic features
- Configuration parameters

i. System Integration and Testing

HW/SW Integration tests	HIL Simulation Interface tests Fault injection Diagnostic coverage
-------------------------	--

System Integration tests	Sensors-ECU-Actuators integration Mode switching Performance under faults
Vehicle Integration tests	Testing the item in the full vehicle Interface with power supply, buses, external ECUs Environmental & real-driving behavior

j. Safety Validation

Xác nhận Safety Goals được đạt ở mức xe:

- Thử nghiệm tình huống nguy hiểm
- Fault injection trên xe thật
- Long-term operational tests
- Validate assumptions trong FSC/TSC

k. Release for production

- • TSC (Technical Safety Concept)
- • TSR (Technical Safety Requirements)
- • System Architecture Design Spec
- • HSI Specification
- • System Integration & Test Strategy
- • System Verification Report
- • Safety Validation Report
- • Safety Analysis Report

5. Product Development at Hardware Level

a. Objective

How to develop hardware so that it meets functional safety requirements (FSR).

It focuses on:

- Translating Technical Safety Requirements (TSR) → Hardware Safety Requirements (HSR)
- Designing safe hardware architecture
- Analyzing random hardware failures
- Calculating hardware safety metrics
- Verifying and testing hardware
- Ensuring hardware achieves its assigned ASIL

b. Hardware Safety Requirements (HSR)

HSR are the technical safety requirements allocated to hardware.

They include:

- Electrical safety (overcurrent, short circuit, voltage faults)
- Hardware fault detection and monitoring
- Triggering safe state via HW

- Latent fault detection requirements
- Timing constraints
- Required diagnostic coverage

c. Hardware Architectural Design

Implement all allocated HSR (sensors, actuators, microcontrollers, power supply, communication interfaces)

Consider:

- Diagnostic capability
- Independence between safety-related channels
- Fault containment
- Simplicity and verifiability
- Separation between safety and non-safety elements

Include hardware safety mechanisms:

- Redundancy
- Monitoring circuits
- Watchdog timers
- Voltage / current monitoring
- Memory ECC
- Clock monitoring

d. Hardware Safety Analysis

Random Hardware Failure Analysis

Performed using:

- FMEDA (Failure Modes, Effects and Diagnostic Analysis)
- FTA (Fault Tree Analysis)
- DFA (Dependent Failure Analysis)

Hardware Safety Metrics

SPFM — Single Point Fault Metric	Measures coverage against single-point failures.
LFM — Latent Fault Metric	Measures coverage against latent dual-point failures.
PMHF — Probabilistic Metric for random Hardware Failures	Maximum hardware failure rate that violates a safety goal: • ASIL D $\rightarrow \leq 10^{-8} / \text{h}$ • ASIL C $\rightarrow \leq 10^{-7} / \text{h}$ • ASIL B $\rightarrow \leq 10^{-7} / \text{h}$ • ASIL A $\rightarrow$ no quantitative requirement

e. Hardware Design Verification

- Schematic review
- PCB layout review
- Simulation

- Timing/worst-case analysis
- Checking diagnostic paths
- Checking independence and redundancy

f. Hardware Integration Testing

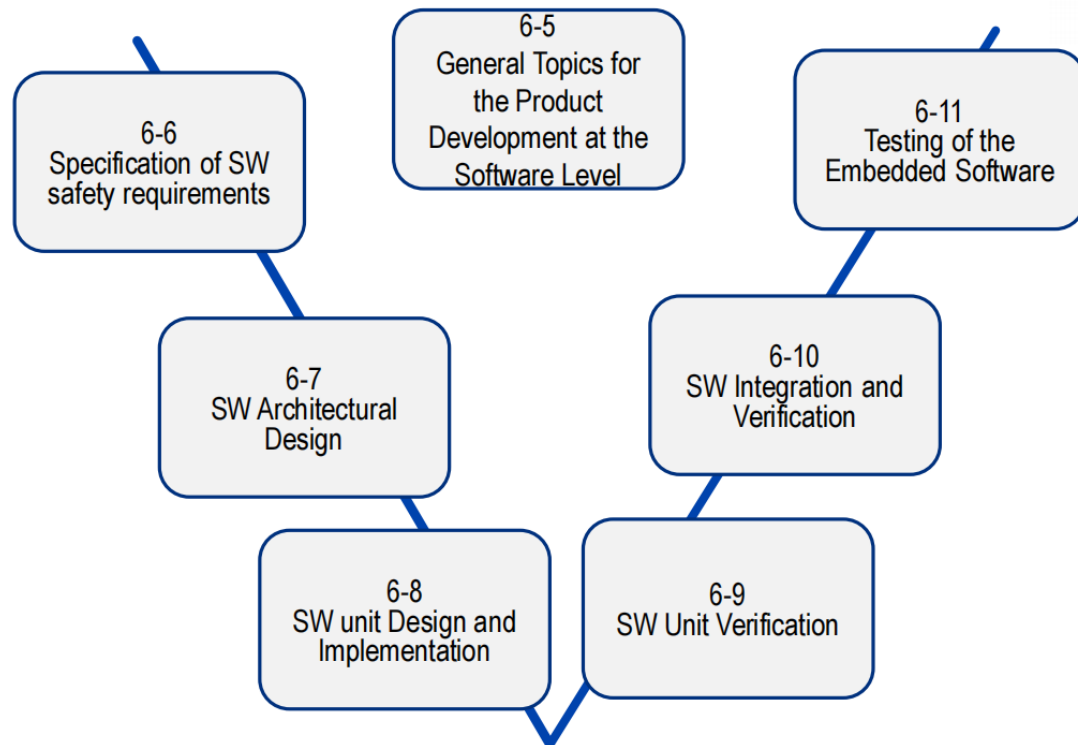
Testing hardware in real conditions:

- Bring-up tests
- Interface tests (sensor ↔ ECU ↔ actuator)
- Fault injection tests
- Environmental tests (temperature, vibration, EMC/EMI)
- End-of-Line (EOL) tests

g. Release for production

- Hardware Safety Requirements Specification
- Hardware Architectural Design Specification
- FMEDA report
- Dependent Failure Analysis (DFA) report
- SPFM / LFM / PMHF calculations
- Hardware verification report
- Hardware integration test report

6. Product Development at Software Level Sub-phase



a. Cover

Deriving Software Safety Requirements (SSR)
Designing the software architecture
Designing & implementing software units
Performing verification at all levels
Ensuring ASIL compliance through confirmation measures

b. Software Safety Requirements (SSR)

SSR are derived from TSR and specify:

- Timing constraints
- Safety mechanisms in software
- Error detection / error handling
- Fault tolerance in software
- Data integrity & interface requirements
- Mode switching & safe state behaviors

SSR must be verifiable, consistent, complete, unambiguous, and traceable.

c. Software Architectural Design

Implement the SSR: Allocate SSR to software components and interfaces.

Define:

- Static architecture (components, interfaces, data types, shared resources)
- Dynamic behavior (execution order, concurrency, scheduling, data flow)

Ensure:

- Modularity & low coupling
- Clear separation of safety-related / non-safety elements
- Freedom from interference (FFI)
- Robustness to faults (e.g., memory protection, timing protection)

Provide software safety mechanisms:

- Range checks / plausibility checks
- Redundant variables
- Timing monitoring
- Watchdog supervision
- Error-state handling

d. Software Unit Design & Implementation

Design & implementation must meet:

- Coding guidelines (e.g., MISRA C, AUTOSAR C++)
- Defensive programming
- Robust data handling
- Error detection (run-time checks)
- Avoiding undefined behavior
- Deterministic execution

ASIL D → highest rigor in design and coding rules

ASIL A → lowest level of rigor

e. Software Unit Verification

Verification must ensure each unit matches the specification through:

- Static analysis
- Code reviews
- Control flow and data flow analysis
- Unit tests (statement/branch/MC/DC coverage depending on ASIL)

Required structural coverage:

- ASIL D → MC/DC
- ASIL C → branch coverage
- ASIL B → statement coverage
- ASIL A → no strict structural requirement

f. Software Integration & Testing

Software units → integrated into components → integrated into final software.

Testing must verify:

- Correct interaction between units
- Correct execution order and timing
- Proper handling of faults and edge cases
- Absence of unintended functionality
- Data flow consistency
- Behavior in all operating modes

Black-box + white-box testing required depending on ASIL.

g. Verification of Software Requirements & Architecture

Performed through:

- Reviews
- Static analysis
- Simulation
- Model checking (optional)
- Requirements traceability analysis
- Walkthroughs

Goal: ensure architecture + SSR fulfill safety & performance needs.

h. Software Release

You must confirm:

- All SSR implemented and verified
- Software quality and safety criteria met
- No blocking bugs or safety anomalies
- Traceability: SSR → architecture → units → tests
- Configuration integrity maintained
- Work products completed and signed off

i. Release for production

- Software Safety Requirements Specification
- Software Architectural Design Specification
- Software Unit Design Specification
- Source Code (according to coding rules)
- Unit Test Reports
- Integration Test Reports
- Requirement Traceability Matrices
- Software Release Report

7. Production, Operation, Service and Decommissioning

a. Objective

Ensures that functional safety is maintained AFTER development, throughout:

- Production
- Vehicle operation
- Maintenance (service & repair)
- Decommissioning (end of life)

It ensures that safety is not lost due to manufacturing errors, incorrect servicing, or unsafe end-of-life handling.

b. Production Planning

Consider:

- Assembly & manufacturing instructions
- Safety-relevant process parameters (reflow temperature, torque values, tolerances)
- Handling and storage conditions
- Correct configuration and flashing of ECU software
- Lessons learned from past production
- Process capability, tools, equipment, test rigs
- Personnel competence

c. Production Plan Content

A production plan must include:

- Process flow & step-by-step instructions
- Production tools & equipment
- Traceability measures (e.g., labeling, serial numbers)
- Rework instructions
- Measures to control safety-related characteristics
- Procedures for loading the correct SW version + calibration data

PFMEA (Process FMEA) must be used to identify foreseeable production failures and implement countermeasures.

d. Operation, Service, and Repair

How the vehicle will be safely used and maintained.

Must include:

- Maintenance & repair procedures
- User information for safe operation (user manual)
- Requirements for decommissioning
- Requirements for rescue/emergency services
- Warning & degradation strategies
- Field monitoring process (collect field failures)
- Handling conditions during service
- Allowed configurations of HW/SW during repair
- Personnel competence

e. Service (Maintenance/Repair) Instructions

Service instructions must specify:

- Procedures, methods, diagnostic routines
- Required tools and equipment
- Safety precautions
- Steps to avoid introducing hazards during repair
- Allowed configurations of hardware/software
- Calibration or flashing procedures for ECUs
- Verification steps after service (e.g., sensor alignment)

f. Decommissioning (ngừng)

Instructions must ensure safe end-of-life handling:

- Removing or disabling safety-related elements
- Avoiding hazards from residual energy, chemicals, or powered components
- Safe disposal of batteries, airbag modules, pyrotechnic devices
- Ensuring no unintended activation occurs

g. Release for production

- Production plans
- Service/maintenance instructions
- User manual (safety-relevant parts)
- Decommissioning instructions
- Field monitoring reports
- Records of safety-related production characteristics

8. Supporting Processes

a. Interfaces in Distributed Development (8-5)

When multiple companies (OEM ↔ Tier-1 ↔ Tier-2) share development:

- A Development Interface Agreement (DIA) must be created

- Defines responsibilities (RASIC), safety managers, outputs & milestones
-
- Ensures each party's safety activities are aligned
- Safety assessments at supplier side must be shared with customer
- After SOP, suppliers must provide evidence of maintained process capability

b. Specification & Management of Safety Requirements

- Safety requirements must be:
- Hierarchical
- Complete
- Atomic
- Internally and externally consistent
- Feasible
- Verifiable
- Unambiguous
- Maintainable
- Traceable across all phases

Tools like DOORS, Polarion, CodeBeamer are recommended.

c. Configuration Management (8-7)

Ensures integrity of:

- Safety work products
- Requirements
- SW/HW versions
- Calibration data
- Must track:
- Version IDs
- Change history
- Dependencies
- Baselines
- Release configurations

d. Change Management

- Any change must undergo:
- Impact analysis
- Documentation update
- Re-verification if needed
- Approval by safety responsible roles
- This prevents systematic failures after modifications.

e. Verification

Checking work products against their specifications:

Includes: Reviews, Analyses, Simulation, Test results

Independence depending on ASIL

f. Documentation Management

Ensures:

- All safety-related documents are controlled
- Correct versions are used
- Proper storage, access control, and traceability

g. Confidence in the Use of Software Tools

- Perform Tool Impact Analysis (TIA)
- Assign Tool Confidence Level (TCL)
- Perform tool qualification when needed

h. Qualification of Software Components

For reused or pre-existing SW components, must prove:

- No safety-impacting behavior
- Sufficient verification evidence
- Robustness under all operating conditions

i. Proven-in-Use Argument

Allows reuse of existing components based on field data.

Requires:

- Sufficient operating hours
- Low incident rate
- Similar operating conditions
- Controlled configuration history

j. Work Products

- DIA (Development Interface Agreement)
- Safety requirement management records
- Configuration & change management records
- Verification reports
- Tool qualification reports
- Proven-in-use reports
- Hardware/SW evaluation reports

9. Automotive Safety Integrity Level (ASIL)-oriented and Safety-oriented Analyses

a. Objective

Its focus is on identifying, evaluating, and mitigating:

- Systematic faults
- Random hardware faults (Part 5)

- Dependent failures / common cause failures (CCF)
- Safety violations due to unintended behavior

b. Four major analysis families

1. Hazard Analysis & Risk Assessment (HARA)
2. Safety-Oriented Analysis During Development (Functional Safety Analysis, Technical Safety Analysis)
3. Dependent Failure Analysis (DFA): 2 kinds: Common Cause Failures và Cascading Failures
4. Safety Analyses for Random Hardware Failures

c. ASIL Decomposition

Splitting a higher ASIL requirement into two or more lower-ASIL elements that provide equivalent safety.

Example: ASIL D requirement → ASIL B/B + ASIL B/B (independent channels)

Conditions:

- Sufficient independence
- No single failure leads to violation
- Proper logging, monitoring, and diagnostics
- Architecture must support separation
- ASIL decomposition helps reduce development effort while preserving safety.

d. Analysis of Unintended Functionality

e. Verification of Safety Analyses

f. Work product

- HARA documentation
- Dependent Failure Analysis (DFA) report
- CCF analysis
- ASIL decomposition rationale
- Safety analysis reports (functional + technical)
- FTA, FMEDA summaries
- Documentation of assumptions, rates, architectures

10. Guidelines on ISO 26262

a. Main Points:

Provides examples, explanations, rationale, and elaboration for all Parts 2–9.

Helps organizations interpret requirements correctly.

Gives guidance on:

Hazard analysis examples

Safety culture, organization setup
Safety management best practices
Safety analysis methods (FMEA, FTA, FMEDA, DFA)
Software and hardware development patterns
Verification & validation approaches

b. SEooC (Safety Element out of Context) in detail:

- How to create it
- Required assumptions
- Verification strategy
- Contains many non-mandatory examples to clarify ambiguous areas.
- Used as a reference during audits and training.

Purpose: Help understand how to implement the other parts correctly.

11. Guidelines on Application to Semiconductor

Main Points:

- Defines how ISO 26262 applies to **semiconductor-level development**.
- Extends Part 5 (hardware) and Part 6 (software) with chip-specific rules.
- Covers:
 - MCU architecture safety mechanisms
 - Lockstep cores, ECC memory, BIST, watchdogs
 - Diagnostic coverage for semiconductor components
 - Failure rate modeling for silicon
 - On-chip redundancy and partitioning
 - CCF inside chips (shared clock, PLL, power supply)
- Provides guidelines for:
 - IP blocks
 - Safety islands
 - Mixed-signal components (ADC, PMIC, IO drivers)
- Used by Tier-1 & semiconductor vendors (NXP, Renesas, Infineon...).
- Includes rules for **ASIL decomposition inside an MCU**.

Purpose: Ensure chips can be developed, evaluated, and qualified for automotive safety.

12. Adaptation of ISO 26262 for Motorcycles

Main Points:

- Tailors ISO 26262 to motorcycle-specific behavior and risks.
- Adjusts **HARA** to motorcycle scenarios:
 - Rider balance
 - Lean angle
 - Single-track stability
 - High exposure risks
- Defines safety for:
 - ABS, TCS, engine control
 - Brake-by-wire, ride-by-wire

- Advanced rider assistance systems (ARAS)
 - Adjusts ASIL determination because:
 - Riders have less physical protection
 - Vehicle stability is more sensitive
 - Provides simplified processes for **low-complexity ECUs**.
 - Includes special rules for:
 - Service and maintenance of motorcycles
 - Functional modifications by end users (common in bikes)
 - Tailors V&V (verification & validation) for motorcycle dynamics.
- Purpose:** Make ISO 26262 practical and realistic for motorcycle manufacturers.